Rawhatur Rabbi

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Overview — Curious and research-driven professional with a strong foundation in Artificial Intelligence. My work centers on developing efficient and interpretable deep learning systems for real-world problems, grounded in scientific rigor and open collaboration. I am motivated to pursue advanced research at the doctoral level, with the long-term goal of contributing to academia through research, teaching, and mentorship while advancing human-centered AI and computational innovation.

Research Interests

Resource-Constrained and Efficient Deep Learning • Representation Learning • Neuromorphic and Spiking Neural Networks • Real-Time Perception and Activity Recognition • Human-Centered and Trustworthy AI Systems • AI for Healthcare • Vision-Language Models (VLMs) • Multimodal Learning

Research & Academic Experience

Adjunct Lecturer (Part Time) **Jan 2026 – Present**
Department of Computer Science and Engineering, Presidency University

- Deliver lectures and conduct laboratory sessions for Object-Oriented Programming (OOP), covering core concepts including classes, inheritance, polymorphism, encapsulation, and abstraction in Java.
- Design course materials, assignments, and lab exercises to reinforce theoretical concepts through hands-on programming practice.
- Evaluate student performance through assessments and provide structured feedback to support student learning outcomes.

Research Assistant (Full Time) **Jan 2026 – Present**
Samsung R&D Institute Bangladesh (SRBD) – Collaborative Project, BRAC University

- Conducting research on stop-motion video rendering pipelines, contributing to the development of efficient frame synthesis and temporal consistency techniques.
- Developing and evaluating deep learning models for video segmentation, enabling precise scene decomposition across complex video sequences.
- Implementing object tracking algorithms to maintain robust identity association and trajectory estimation across frames in dynamic scenes.
- Investigating amodal and modal instance segmentation methods to recover complete object representations under occlusion in real-world video data.

Research Assistant (Full Time) **Sep 2023 – Nov 2025**
Department of Computer Science and Engineering, BRAC University

- Led technical development within an RSGI-funded research project on real-time fall detection and activity recognition, including dataset design and system implementation.
- Developed and deployed real-time human activity recognition pipelines integrating deep learning and computer vision, achieving **90% accuracy** on custom motion datasets.
- Engineered scalable data processing and real-time streaming infrastructure to support low-latency model inference and deployment.
- Co-authored multiple peer-reviewed journal and conference publications and contributed to successful grant proposals in machine learning and computer vision.
- Mentored **30+ undergraduate thesis students** and supported interdisciplinary research through data engineering and analytics across academic units.

Undergraduate Teaching Assistant **Jan 2022 – Apr 2022**
School of Data & Sciences, BRAC University

- Provided personalized one-on-one instruction to address individual learning needs.
- Simplified complex math and computer science concepts for better comprehension.
- Tracked student progress and tailored tutoring sessions accordingly.
- Provided coding guidance, helping with algorithm design and debugging.
- Created and shared study materials, including practice problems and solutions.

Industry Experience

Software Engineer Trainee (AI & ML) **Apr 2023 – Sept 2023**
Medina Tech Ltd. (now Genius Farms Limited)

- Enhanced datasets through data augmentation, diversifying data by 30%, and performed in-depth data analysis, leading to a 20% improvement in model performance.
- Implemented geocode visualization methods, identifying location-based trends, and generated reports.
- Collaborated on backend development using Django and Odoo, contributing to the seamless functionality of web applications and databases.
- **Technical Skills:** Python, Keras, Tensorflow, NumPy, Pandas, Linux, PostgreSQL, Django, Odoo, Git.

QA & Automation Tester **Jul 2021 – Jan 2024**
Master Wizr, Oslo, Norway

- Developed and executed comprehensive test plans and cases for software applications to ensure alignment with system specifications.
- Automated repetitive testing tasks using tools like Selenium and Postman, reducing manual testing time by 50%.
- Identified, documented, and reported bugs, tracking them through to resolution to maintain software quality standards.
- **Soft Skills:** Selenium, Pytest, Jira.

Education

BRAC University, Dhaka, Bangladesh **Apr 2024 – Present**
MSc in Computer Science and Engineering

- CGPA: 3.83/4.00 (ongoing)
- Worked as a Graduate Research Assistant
- Awarded Merit-Based Scholarship
- **Courses:** Symbolic Machine Learning, Advanced Artificial Intelligence, Information Assurance and Human Factor, Pattern Recognition, Advanced NLP
- **Master's Thesis:** Real-Time Fall Detection for Edge Devices Using Distilled Video Representations *(In progress)*

BRAC University, Dhaka, Bangladesh **Jan 2019 – Dec 2022**
BSc in Computer Science and Engineering

- CGPA: 3.62/4.00 (High Distinction)
- Worked as a Student Tutor
- **Undergraduate Thesis:** Lightweight Custom CNNs for Corn Leaf Disease Classification [Pdf]

Notre Dame College, Dhaka, Bangladesh **Mar 2016 – Aug 2018**
Higher Secondary School Certificate

- GPA: 5.00/5.00

Publications

Journal Articles

1. Datta, J., **Rabbi, R.**, Saha, P., et al. **Deep Representation Learning using Layer-wise VICReg Losses**. ***Scientific Reports*** (Q1 Journal), 2025. [paper]

Conference Papers

2. **Rabbi, R.**, Afaf Sarwar, F., Islam, M. F., Ahmed, T., Rudro, S., Rahman, M. M., Manab, M. A., and Mukta, J. N. **Contrast then Confidence (C^2): Contrastive Pretraining for Uncertainty-Aware Out-of-Distribution Detection in Satellite Imagery.** ***GeoCV Workshop, IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)***, 2026. [Accepted]
3. Datta, J., **Rabbi, R.**, Tonmoy, M. S. I., Saha, P., and Mourning, C. **POM-NeRF: Perception-Oriented Modification of Neural Radiance Fields for Enhanced 3D Volumetric Rendering from Multi-View Images**. ***Proc. VISAPP***, Marbella, Spain, 2026. **(CORE Rank B)** [Accepted]
4. Rafin, N. I., **Rabbi, R.**, Tonmoy, S. I., Alam, M. G. R. **Optimizing Building Energy Consumption with Genetic Algorithms and Explainable AI**. ***Proc. ICCIT***, Cox's Bazar, Bangladesh, 2025. [Presented]
5. Tasnim, H., Joy, A. D., Dutta, A., **Rabbi, R.**, Zereen, A. N. **Temporal Fusion of CNN–LSTM Networks for Vision-Based Fall Detection Using Anatomical Keypoints**. ***Proc. ICCIT***, Cox's Bazar, Bangladesh, 2025. [Presented]

6. Ahmed, N., Hasan, M. R., Shakib, F. H., Rabbani, H., **Rabbi, R.**, Khan, S. T., Zereen, A. N. **Human Fall Detection Through Velocity-Driven Temporal Deep Learning**. ***Proc. ICCIT***, Cox's Bazar, Bangladesh, 2025. [Presented]
7. Datta, J., **Rabbi, R.**, Rafin, N. I., Alam, M. G. R. **Peer-Guided Optimization: Incorporating Collaborative Learning into Stochastic Optimization in Machine Learning**. ***Proc. ECCE***, Chittagong, Bangladesh, 2025. [paper]
8. Datta, J., Sarwar, F. A., **Rabbi, R.**, Saha, P., Alam, M. G. R., Uddin, J., Zereen, A. N. **Parameter-Efficient Image Classification with Convolutional Spiking Neural Networks via Fast Sigmoid Surrogate Gradient Descent**. ***Proc. ECCE***, Chittagong, Bangladesh, 2025. [paper]
9. **Rabbi, R.**, Datta, J., Ahmed, S., Zereen, A. N. **Wildfire Prediction using Convolutional Kolmogorov–Arnold Networks**. ***Proc. ICECE***, Dhaka, Bangladesh, 2024. [paper]
10. Arefin, M. Y., **Rabbi, R.**, Turna, I. F., Zannat, Z., Karim, D. Z. **Lightweight Custom CNNs Take on Corn Leaf Disease: Deep Learning for Plant Pathology**. ***Proc. ICECCME***, Canary Islands, Spain, 2023. [paper]

Under Review

1. **SE-CCT: Compact Convolutional Transformer with Channel Attention for Deepfake Image Detection**. Submitted to ***Egyptian Informatics Journal*** *(As first author)*
2. **ACEPNet: An Enhanced CNN-Attention Framework for Predicting National Per Capita Electricity Consumption using Nighttime Satellite Imagery and Geo-Demographic Data**. Submitted to ***IEEE Transactions on Geoscience and Remote Sensing***.

Projects

AgriBuddy
AI Hackathon, Bangladesh Innovation Conclave, BRAC University

– Built a multilingual AI agricultural assistant that improves farmer decision-making through context-aware recommendations tailored to local Bangladeshi conditions.
– Demonstrated improved recommendation relevance and user engagement in field evaluations using an agentic RAG framework with Bangla embeddings. [Technical Report]

Dr. Chashi
Medina Tech Ltd. (now Genius Farms Limited)

– Improved reliability of digital farming insights by training deep learning models on real field-collected datasets.
– Increased model robustness and practical usability through systematic data preprocessing and close collaboration with agricultural domain experts.

Skills

– **Programming & Systems:** Python, C++, Java, JavaScript/TypeScript; Linux, Bash/Zsh, Git
– **ML Framework:** PyTorch, TensorFlow, Keras
– **NLP & LLMs:** Transformer models, Retrieval-Augmented Generation (RAG), LangChain-based pipelines, prompt engineering, vector databases, and embedding-based retrieval
– **Backend & APIs:** Django, Flask, FastAPI, Node.js, Express.js
– **Data & Storage:** PostgreSQL, MySQL, MongoDB, Firebase; FAISS, ChromaDB
– **Deployment & MLOps:** Docker, MLflow, DVC; edge deployment on NVIDIA Jetson
– **Frontend & Mobile:** React.js, HTML/CSS, Flutter

Awards and Achievements

– Finalist, Agriculture, AI Hackathon, Bangladesh Innovation Conclave (2025)
– Awarded merit-based scholarships for academic excellence in both undergraduate (BSc) and graduate (MSc) studies at BRAC University.
– Finalist, Line Follower Robot, RUET CSE FEST, RUET (2022)
– Finalist, Hackathon of Joyjatra'21 National Fest, Bangladesh (2021)
– 2nd Runner-up, Intra-University Programming Contest, BRAC University (2020)
– Vice Chancellor's (VC) list, Listed for outstanding academic performance, BRAC University (2019)
– Life Time Membership, Notre Dame Science Club, NDC (2018)
– 2nd Position, Web Designing Competition, Notre Dame Annual Science Festival 2014 & 24th GKC, NDC (2014)

References

References will be provided upon request.